

SECURE YOUR JOB WITH AI

MASTER GEN AI & AGENTIC AI

In just 7 Days

For Working Professionals Across All Industries

7 Days | 6 Hours/Day | 42 Hours Total

Python → Deep Learning → Gen AI → Agentic AI → Real-World Applications



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NOW**



AI is no longer a future topic — it is happening right now across every industry, and the professionals who understand it will lead the next decade. This course is your fast-track to becoming one of them.

In just 7 days, you will go from zero to building real AI systems that organisations are deploying today — systems that automate complex workflows, serve customers 24/7, qualify leads intelligently, process documents in minutes, and make decisions that used to take days. You do not need to be a programmer or data scientist to start. Every concept is taught step by step, with real-world examples drawn from banking, e-commerce, sales, HR, and operations.

By the end, you will know how AI thinks, how to direct it, and how to apply it to the real challenges your organisation faces — whether you are in operations, sales, technology, compliance, or leadership. You will walk away with the confidence to lead AI conversations, evaluate AI solutions, and build production-ready systems that solve problems across any domain

Course Overview

Day	Theme	Hours
Day 1	Python & Deep Learning Foundations	6 hrs
Day 2	Transformers, LLMs & Prompt Engineering	6 hrs
Day 3	Agentic AI Foundations & RAG Systems	6 hrs
Day 4	Agentic Frameworks & Multi-Agent Systems	6 hrs
Day 5	Fine-Tuning, Multimodal AI & Agent Memory	6 hrs
Day 6	Safety, Evaluation, MCP & Observability	6 hrs
Day 7	Real-World Projects & Capstone	6 hrs

DAY 1

Python & Deep Learning Foundations

Duration: 6 Hours | Level: Foundations

Build the technical base — Python programming, data handling, neural networks, and NLP fundamentals that power every modern AI system.

Session 1 – Python Fundamentals

Topics Covered

- Variables, Data Types, Control Flow, Functions, OOP
- Lists, Dictionaries, Sets, Comprehensions
- File I/O, Exception Handling, Modules



Session 2 – NumPy & Pandas for Data Analysis

Topics Covered

- NumPy Arrays, Broadcasting, Vectorized operations
- Pandas DataFrames – slicing, filtering, groupby, merge
- Time-series handling, rolling windows, aggregations



Session 3 – Data Visualization

Topics Covered

- Matplotlib – line, bar, scatter, histogram plots
- Seaborn – heatmaps, pair plots, distribution plots
- Storytelling with data



Session 4 – Neural Networks & Deep Learning

Topics Covered

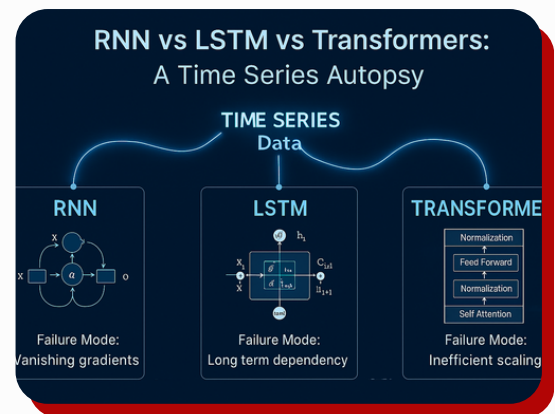
- Perceptrons & Multi-Layer Perceptrons (MLPs)
- Activation Functions: ReLU, Sigmoid, Tanh, Softmax
- Forward & Backward Propagation
- Loss Functions & Optimizers: SGD, Adam, RMSprop
- Regularization: Dropout, L1/L2



Session 5 – RNNs & LSTMs

Topics Covered

- Sequential data processing, Hidden states, Temporal dependencies
- Vanishing/Exploding Gradient Problem
- LSTM architecture: Forget, Input, Output gates
- Bidirectional LSTMs
- CNNs: Convolutional layers, pooling, feature maps



Session 6 – NLP Fundamentals (Pre-Transformer Era)

Topics Covered

- Text preprocessing: tokenization, stop words, stemming, lemmatization
- Bag of Words (BoW), TF-IDF
- Word Embeddings: Word2Vec, GloVe
- Encoder-Decoder Seq2Seq architecture, limitations leading to attention



DAY 2 Transformers, LLMs & Prompt Engineering

Duration: 6 Hours | Level: Intermediate

Understand the architecture behind ChatGPT, Claude, and every modern AI assistant. Master prompts that work in production.

Session 1 – Transformer Architecture Mastery

Topics Covered

- Self-Attention: Query, Key, Value (QKV) matrices
- Scaled Dot-Product Attention
- Multi-Head Attention
- Why Attention beats RNNs for long documents



Session 2 – Positional Encodings & Tokenization

Topics Covered

- Why position matters in transformers
- Sinusoidal positional encoding
- Relative positional encodings: RoPE, ALiBi
- Tokenization: BPE, WordPiece, SentencePiece
- Subword tokenization advantages



Session 3 – LLM Architecture & Variants

Topics Covered

- BERT (Encoder-only) – classification, NER
- GPT (Decoder-only) – generation, completion
- T5 (Encoder-Decoder) – summarization, translation
- Embedding layers, Transformer blocks, LayerNorm
- Context length limitations, KV cache optimization



Session 4 – Sampling, Generation & Alignment

Topics Covered

- Temperature, Top-k, Top-p (nucleus) sampling
- Beam search, Greedy decoding
- RLHF – Reinforcement Learning from Human Feedback
- DPO – Direct Preference Optimization
- Constitutional AI



Session 5 – Prompt Engineering Fundamentals

Topics Covered

- Zero-shot, One-shot, Few-shot prompting
- System, User, Assistant role structure
- Chain-of-Thought (CoT) prompting
- Tree of Thoughts (ToT)
- Self-consistency



Session 6 – Structured Outputs & Prompt Security

Topics Covered

- JSON mode and schema enforcement
- Function calling patterns
- Prompt injection awareness and delimiter usage
- Token optimization, prompt versioning
- LangSmith for prompt debugging



DAY 3

Agentic AI Foundations & RAG Systems

Duration: 6 Hours | Level: Intermediate

Move from chat to action – build AI agents that can reason, plan, and retrieve information from your organisation's documents.

Session 1 – Agentic AI Fundamentals

Topics Covered

- AI Agents vs Models – key distinctions
- Autonomous vs Semi-Autonomous agents
- Perception-Action Loop: Observe → Think → Act → Feedback
- Autonomy levels and use cases
- Agent limitations and failure modes



Session 2 – Planning, Reasoning & Tool Use

Topics Covered

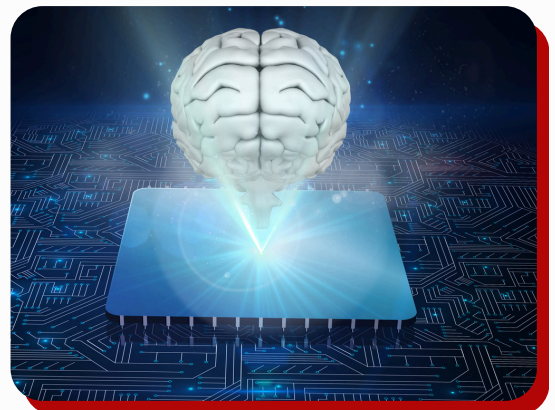
- Task decomposition and goal setting
- Planning algorithms: A*, MCTS
- ReAct paradigm (Reasoning + Acting)
- Plan-and-Execute pattern
- Tool use: function calling, web browsing, code execution



Session 3 – Agent Memory Systems

Topics Covered

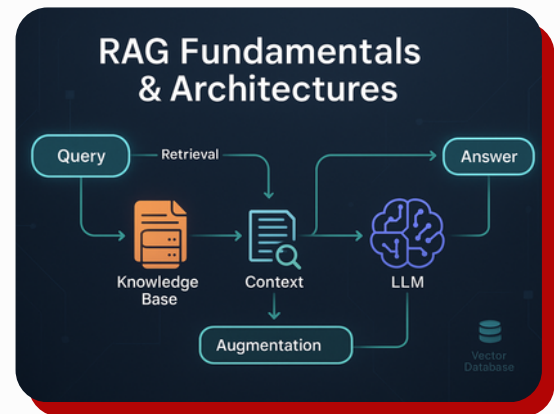
- Working memory (context window)
- Short-term memory: conversation history
- Long-term memory: vector stores, databases
- Episodic vs Semantic memory
- State tracking and session management



Session 4 – RAG Fundamentals

Topics Covered

- RAG architecture: Retriever + Reader + Generator pattern
- Why RAG? Hallucination mitigation
- When to use RAG vs fine-tuning
- Embedding models: OpenAI, Cohere, BGE, E5
- Cosine similarity and retrieval metrics



Session 5 – Vector Databases & Document Processing

Topics Covered

- Vector DBs: FAISS, Pinecone, Weaviate, Chroma, Qdrant, Milvus
- Indexing strategies: HNSW, IVF
- Metadata filtering
- Chunking strategies: fixed-size, semantic, overlap
- Document processing: OCR, PDF parsing, table extraction



Session 6 – Advanced RAG Patterns

Topics Covered

- Hybrid search: dense + sparse (BM25)
- Re-ranking mechanisms
- Hypothetical Document Embeddings (HyDE)
- Query decomposition, multi-query retrieval
- Graph RAG: knowledge graphs + entity linking
- RAG Evaluation: RAGAS, ARES frameworks



DAY 4 Agentic Frameworks & Multi-Agent Systems

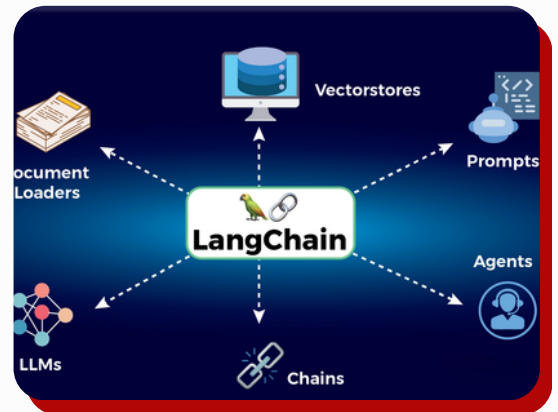
Duration: 6 Hours | Level: Intermediate

Build production-grade AI pipelines using LangChain, LangGraph, AutoGen, and CrewAI.

Session 1 - LangChain

Topics Covered

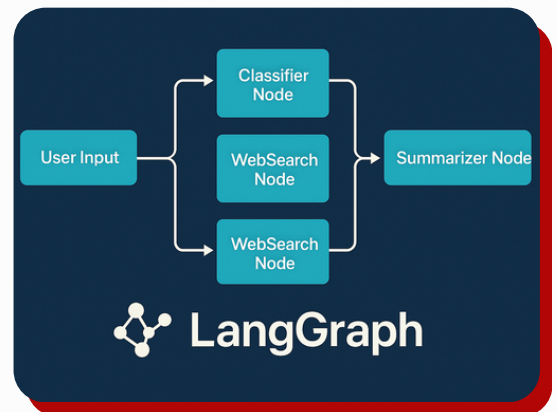
- LCEL (LangChain Expression Language) and chains
- Agents and tools
- Memory management in chains
- Callbacks and tracing
- Custom tool development



Session 2 - LangGraph for Stateful Workflows

Topics Covered

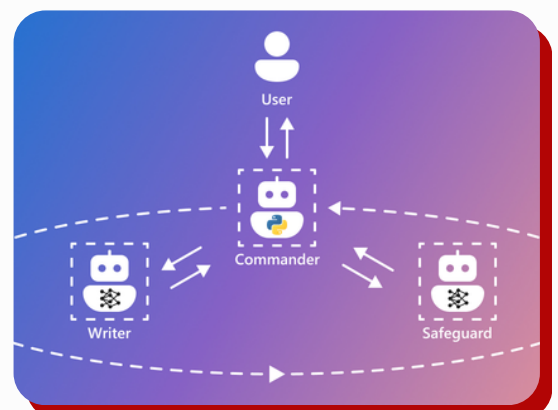
- Graph-based stateful workflows
- State machines and conditional routing
- Checkpointing and persistence
- Human-in-the-Loop (HITL) integration



Session 3 - AutoGen for Multi-Agent Systems

Topics Covered

- Conversable agents and group chat
- Multi-agent orchestration
- Code execution agents
- Human-in-the-loop patterns



Session 4 – CrewAI for Role-Based Agent Teams

Topics Covered

- Role-based agent design
- Task assignment and delegation
- Crew orchestration patterns
- Framework comparison: LangChain vs LangGraph vs AutoGen vs CrewAI



Session 5 – Multi-Agent Architecture & Communication

Topics Covered

- Centralized vs Decentralized control
- Communication protocols: message passing, pub-sub, event-driven
- Agent routing and semantic routing
- Fallback mechanisms and error handling



Session 6 – Hierarchical Agents & Human-in-the-Loop

Topics Covered

- Manager-Worker patterns
- Task decomposition (RACI matrix)
Collaborative vs competitive agents
- Approval workflows and human feedback integration
- Reflection agents and critique-refine loops



DAY 5 Fine-Tuning, Multimodal AI & Agent Memory

Duration: 6 Hours | Level: Intermediate

persistent agent memory

Session 1 - Fine-Tuning Fundamentals

Topics Covered

- When to fine-tune vs RAG vs prompt engineering
- Full fine-tuning vs PEFT
- Data requirements: JSONL format, train/validation split
- Quality over quantity principle



Session 2 - Parameter-Efficient Fine-Tuning (PEFT)

Topics Covered

- LoRA (Low-Rank Adaptation) — how it works
- QLoRA (Quantized LoRA) — memory-efficient training
- Adapter layers and Prefix tuning
- Instruction tuning and SFT (Supervised Fine-Tuning)



Session 3 - Agent Memory & Cross-Session Persistence

Topics Covered

- Multi-tier memory architecture
- Short-term: context window and conversation history
- Long-term: vector stores and database storage
- Episodic memory: event sequence storage
- Semantic memory: fact storage, knowledge graph integration
- State management: SQLite, Redis, PostgreSQL
- State checkpointing, privacy considerations



DAY 6

Safety, Evaluation, MCP & Observability

Duration: 6 Hours | Level: Intermediate

Deploy AI responsibly – guardrails, evaluation frameworks, the Model Context Protocol, and production monitoring.

Session 1 – AI Safety & Content Moderation

Topics Covered

- AI Safety Principles: alignment, robustness, transparency
- Content Moderation: toxicity detection, PII detection
- Input validation and sanitization: length limits, format checks
- Output validation: factuality verification, hallucination detection
- Constitutional AI and value alignment



Session 2 – Prompt Injection Defense & Hallucination Mitigation

Topics Covered

- Understanding prompt injection attacks
- Delimiter-based defenses
- Instruction hierarchy and system message hardening
- Jailbreak prevention: refusal training, multi-layer defenses
- Hallucination Mitigation: Chain-of-Verification (CoVe), confidence scoring, citation requirements



Session 3 – Evaluation Frameworks

Topics Covered

- Evaluation Framework Design: metric selection, benchmark creation
- Offline evaluation: automated metrics (BLEU, ROUGE, F1, embedding similarity)
- Online evaluation: A/B testing, canary deployment, champion-challenger
- LLM-as-Judge: criteria-based evaluation, bias considerations
- RAG-specific: context relevance, answer faithfulness, RAGAS and ARES Agent evaluation: task completion rate, tool usage accuracy



Session 4 – Observability & Monitoring

Topics Covered

- Observability fundamentals: metrics, logs, traces
- Distributed tracing for AI pipelines
- LangSmith platform: trace visualization, dataset management, prompt playground
- Grafana dashboards: metrics visualization and alerting
- Cost tracking, latency analysis, token usage monitoring



Session 5 – Model Context Protocol (MCP)

Topics Covered

- MCP protocol: architecture, MCP vs traditional APIs
- MCP Server Development: tool registration, schema definition
- MCP Client Integration: tool discovery, error handling
- LLM Connectors: connecting LLMs to external systems
- MCP Security: authentication, permission model, rate limiting
- MCP Deployment: containerization, orchestration, monitoring



Session 6 – Red-Teaming & Responsible AI

Topics Covered

- Red-teaming methodology for AI systems
- Adversarial testing and penetration testing
- OWASP Top 10 for LLMs
- Human oversight: approval workflows, audit logging, escalation
- Safety classifiers: multi-stage filtering
- Building audit trails for compliance and governance



DAY 7

Real-World Projects & Capstone

Duration: 6 Hours | Level: Intermediate

Build 3 complete AI systems across different domains – from design to deployment. Portfolio-ready projects you can showcase immediately

PROJECT 1 Loan Application & Processing Agent

Domain: Banking & Financial Services | **Type:** Agentic AI

Frameworks: LangGraph (7-node workflow)

Problem:

Manual loan processing takes 5–7 days, involves 28+ processors, and handles only 600 loans per month with high error rates and poor customer experience.

What You Build:

An end-to-end autonomous loan processing agent – from application submission to disbursement – with no manual intervention for standard cases.

Workflow Nodes:

- Document intake + OCR extraction
- ID authenticity verification
- Credit score analysis
- Debt-to-income ratio calculation
- Property / collateral valuation
- Risk scoring + interest rate determination
- Auto-approve / escalate to human / disburse

Agentic Capabilities:

- Multi-step autonomous decision making
- Human-in-the-loop escalation for edge cases
- Document verification using Vision AI
- Real-time status updates to applicant

Tech Stack: LangGraph, GPT-4o, Qdrant, Unstructured.io (OCR), e-signature API, Streamlit

Real Benchmark: HDFC Bank

Loan approval: 5 days → 2 minutes | Capacity: 600 → 15,000 loans/month (25x) | Default accuracy: 78% → 94%



PROJECT 2 Customer Support Chatbot

Domain: E-commerce / CustomerService | **Type:** Multi-Agent + MCP

Frameworks: LangGraph + MCP

Problem:

Support costs \$1.3 trillion globally. Average wait time is 11 minutes, causing 75% customer abandonment. Agents spend 60% of time on repetitive tasks. Traditional chatbots fail at multi-step problems and cannot take real actions like processing refunds. Support is unavailable 16 hours/day for most businesses.

What You Build:

An autonomous multi-agent support system handling 70% of queries end-to-end — checking orders, processing refunds, updating records, and escalating only when needed.



Multi-Agent System:

- **Triage Agent** – classifies intent, routes to correct agent
- **Order Management Agent** – checks order status, tracking, delivery
- **Product Info Agent** – answers product queries, availability, specs
- **Billing Agent** – handles payments, invoices, refunds
- **Technical Support Agent** – troubleshooting, step-by-step guidance
- **Escalation Agent** – hands off to human with full context

MCP Integration:

Centralized tool registry connecting all business systems— order database, inventory system, payment processor, email service, ticket creation

Decision Logic: Business rules engine for refunds, returns, and escalation thresholds

Tech Stack: LangGraph, GPT-4o-mini, Weaviate, MCP (business system connectors), WebSocket, Streamlit

Key Metrics:

70% queries resolved end-to-end | Response time < 30 seconds | 24/7 availability | Support cost reduced by 60%+

PROJECT 3 Lead Scoring Agent

Domain: Sales & Marketing | | **Type:** Agentic AI + Web Search

Frameworks: CrewAI + Web Intelligence

Problem:

Sales teams waste 60% of their time on unqualified leads – contacting 100 prospects to close just 3 deals. Manual research takes 15–30 minutes per lead with no systematic qualification process, resulting in \$50,000+ wasted effort per sales rep annually. There is no data-driven way to prioritize high-probability prospects.

What You Build:

An autonomous lead qualification agent that researches, scores, and prioritizes prospects – giving sales teams a ranked list of high-conversion leads with personalized talking points ready to go.



Agentic Workflow:

- **Research Agent** – scrapes company website, LinkedIn, news, funding data, tech stack
- **Analysis Agent** – matches lead against Ideal Customer Profile (ICP)
- **Contact Agent** – identifies decision-makers and generates contact information
- **Scoring Agent** – scores lead 0–100 based on conversion probability

Data Sources:

Web search, company websites, LinkedIn, tech stack databases, funding databases, news APIs

Scoring Model Inputs:

- ICP matching (industry, company size, revenue range)
- Buying signals (recent funding, hiring trends, tech adoption)
- Engagement signals (website visits, content downloads)
- Decision-maker accessibility

Output Per Lead:

- Lead score (0–100) with reasoning
- Decision-maker contacts and roles
- Personalized talking points tailored to company context
- Recommended outreach timing and channel

Tech Stack:

CrewAI, GPT-4o, Tavily (web search), Qdrant, LinkedIn API, Streamlit

Key Metrics:

15–30 min manual research → 2 min automated | Sales team focuses only on top 20% leads | Conversion rate improved 3–5x

Capstone: Demo Day

- Present your chosen project end-to-end
- Live demo with real-world scenarios
- Q&A with industry mentors
- Portfolio documentation and GitHub repository setup
- LinkedIn profile optimization for AI roles

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